MODERN METHODS OF GENAI

DashLab

Generating and repairing Geometry Dash levels

A local LLM with external verification and native export

Project defense

Creating levels takes repeated design and playtesting

The creator’s loop

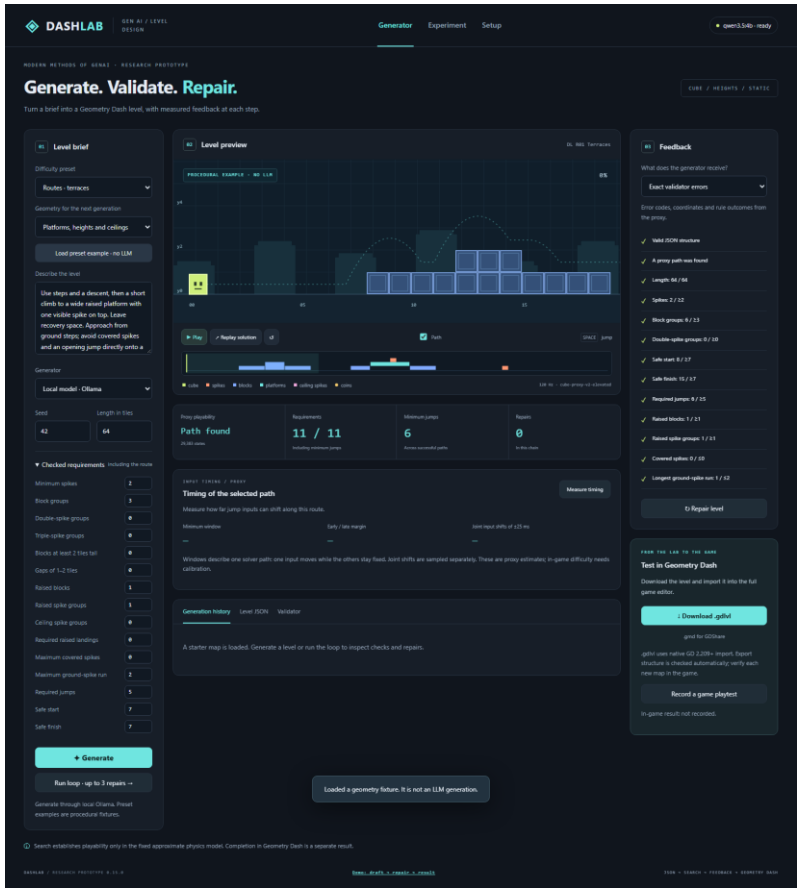
Place obstacles, test transitions, revise the layout.

Each new brief repeats that work

Different requirements need another round of design and testing.

Our goal

Generate and repair a level prototype that the creator can export and edit.



Reduced authoring effort motivates the project. We did not measure time savings.

Qwen chooses patterns, the compiler places objects

Section 1
Preserved during this edit

Section 2
Local repair target

Section 3
Preserved during this edit

Section 4
Preserved during this edit

A fixed library turns pattern choices into coordinates

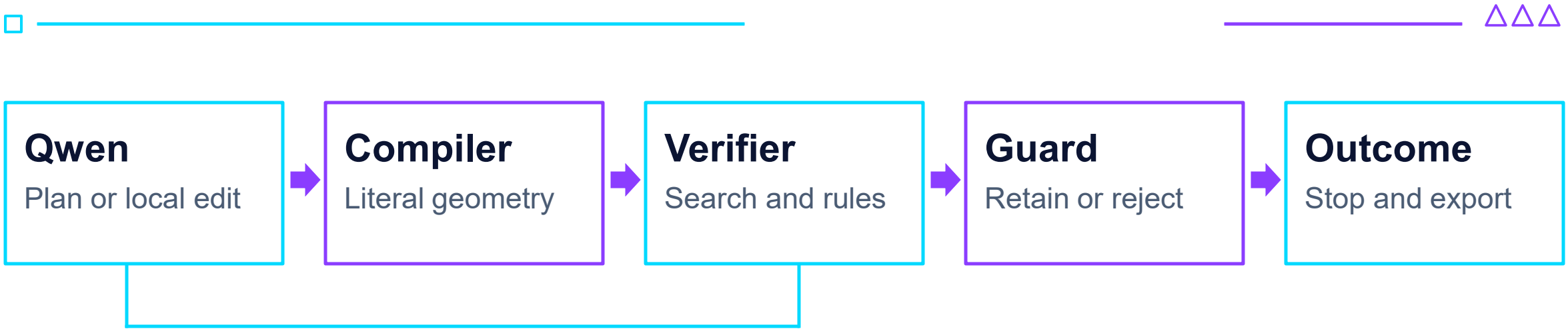
Real M03 section after repair

{"pairs":["tower_triple","triple_pair","stairs"],"extra":"tall_block"}

Four bounded sections. Literal proposals. No hidden compiler repair.

The shared library also makes random repair a necessary control.

An external verifier checks the level



Independent checks

Format and bounds, a full path, and every numeric requirement.

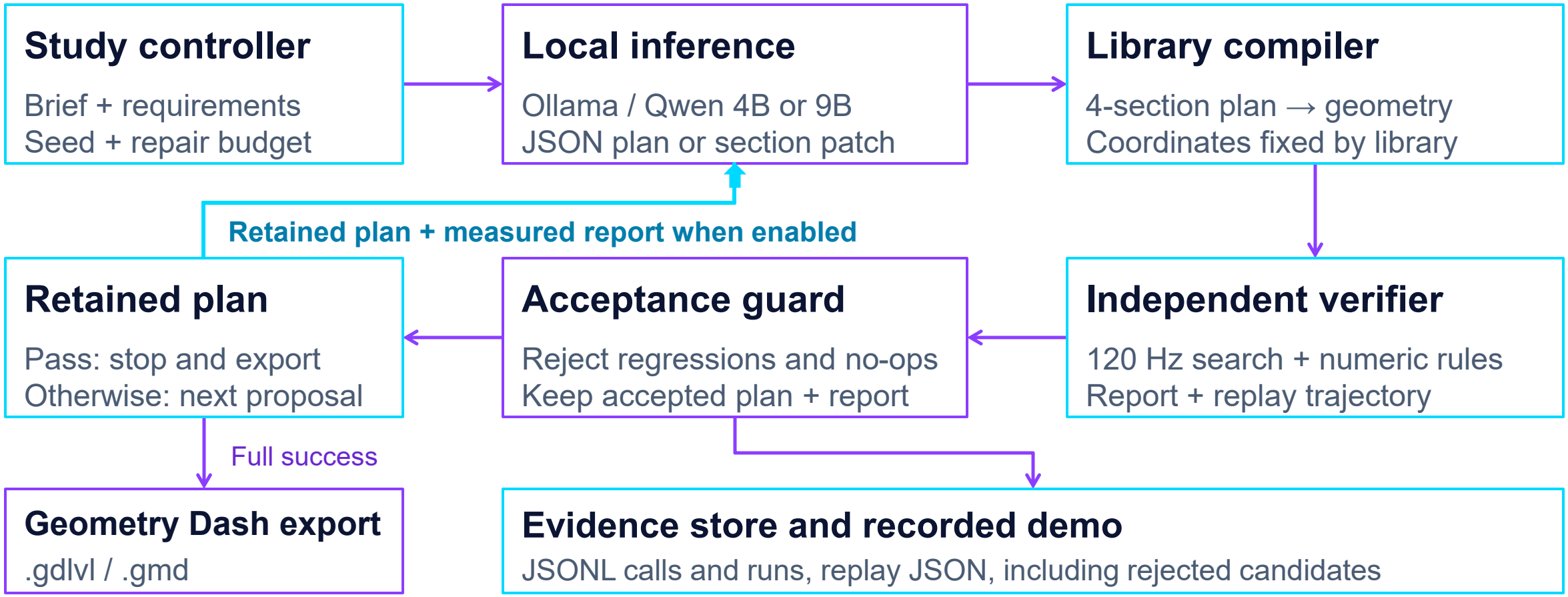
Search in fixed cube physics

120 Hz. Jump or no jump. Keep the lower-jump matching state.

Ollama runs Qwen locally. Search uses approximate physics, not the full game engine.

Project architecture and data flow

Node.js serves the browser UI. A separate CLI runner executes the final study.



Modules: server.js, retained-plans.js, engine.js, incremental-repair.js, gd-export.js

Failed candidates trigger repair

Already passes? Stop with zero repairs.

Propose
Up to 3 repair proposals

Recheck
Measure path and requirements

Retain or reject
Preserve passing checks

Diagnostic text

Measured deficits, failure location and static replacement tradeoffs.

Common guard

Reject lost checks, earlier failure or a larger deficit under the guard rules.

Zero lost checks is a controller property. Static hints are not pretested solutions.

One shared initial plan, six repair conditions

12 briefs × 2 seeds × 2 models × 6 conditions = 288 chains

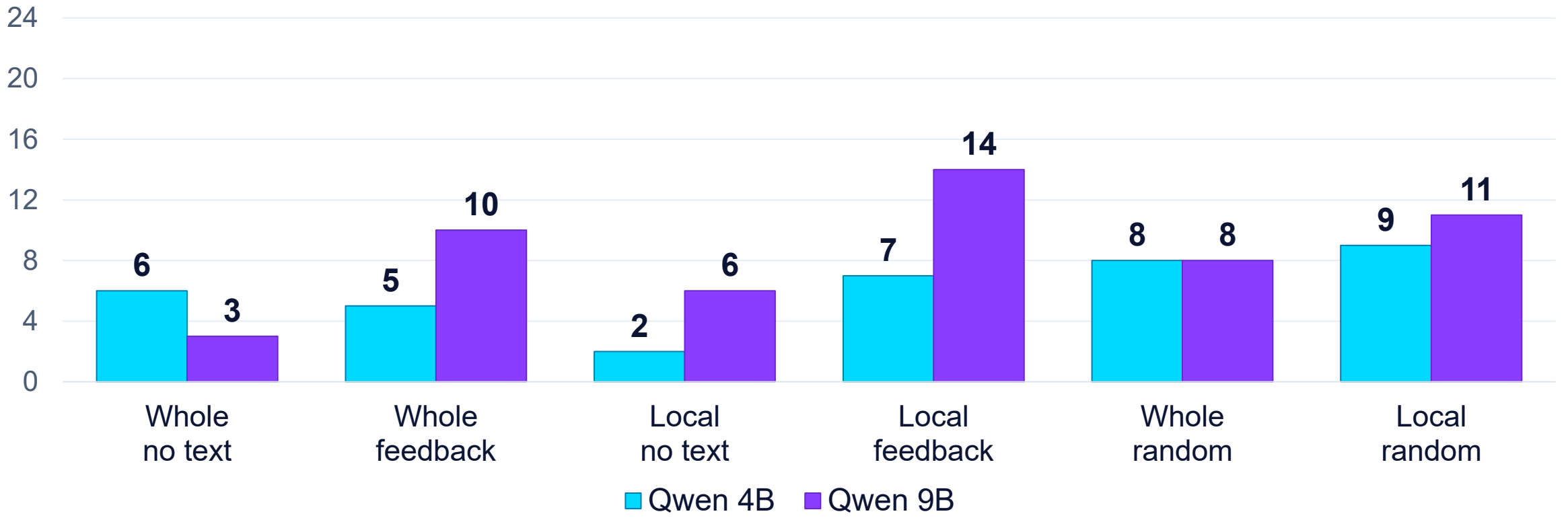
Repair scope	No diagnostic text	Measured feedback	Random repair
Whole plan	Qwen rewrite	Qwen + report	Random rewrite
One section	Qwen replacement	Qwen + report	Random replacement

Qwen3.5 4B and 9B, Q4_K_M
Same library, initial answer, guard and maximum of 3 repairs

12 numeric briefs in 6 existing families. Local no-text control still receives a target and acceptance.

Local feedback helped 9B

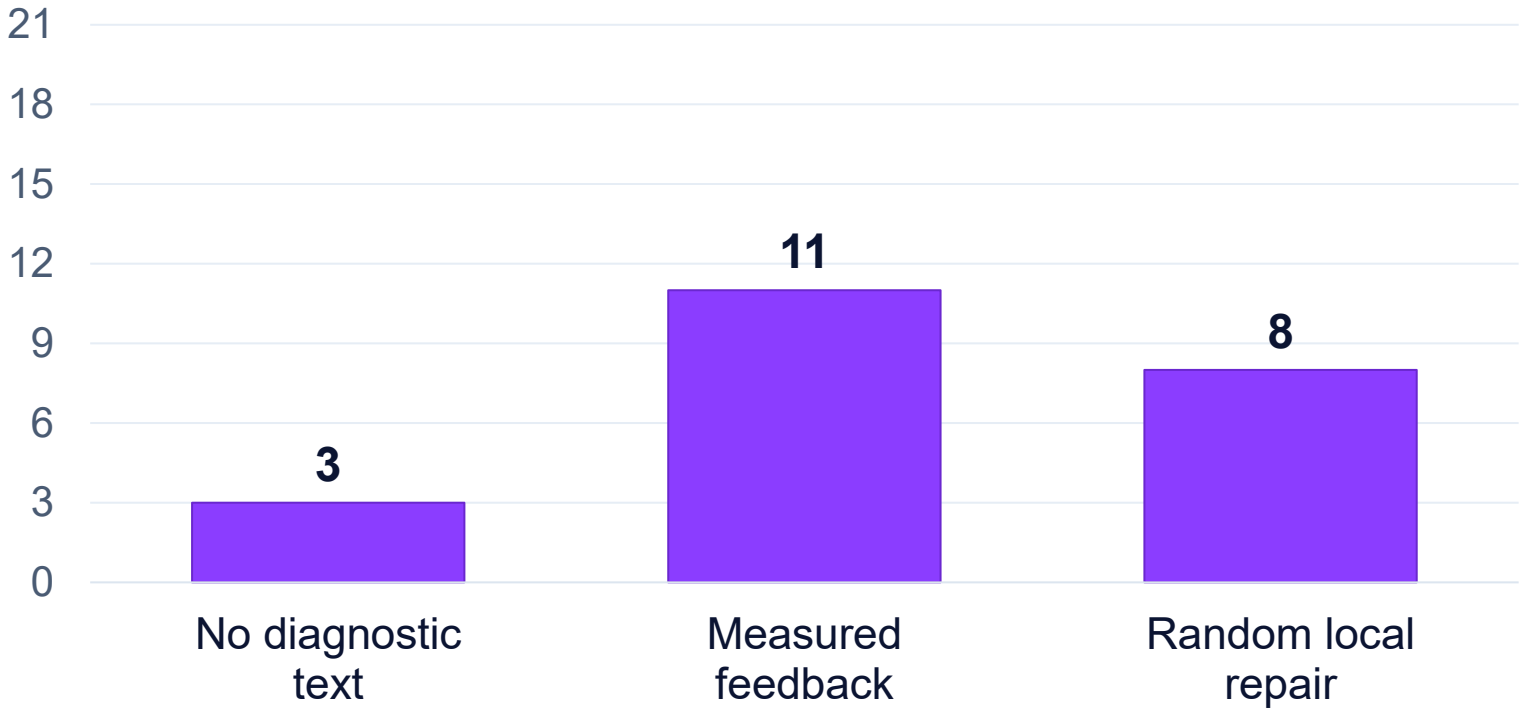
Complete proxy successes out of 24 per condition



9B is secondary and exploratory. Superiority over random repair is not established.

Recovery isolates the repair gain

Qwen 9B: 21 failed starts after excluding 3 shared initial successes



Unchanged proposals

16 / 48

with diagnostic text

44 / 61

without diagnostic text

Chains stop on success, so proposal counts differ. This is not an equal-token-budget comparison.

One local edit turns NO_PATH into PASS

M03, Qwen 9B. Recorded real response and search replay.

Section 2: triple_pair → tower_triple. Other 3 sections unchanged.



Fixed approximate physics. Example selected after outcomes. Browser demo: /repair-demo.html

A path alone does not satisfy the request



Repair 2 has a path, but loses checks

27 spikes < 30 required
9 double patterns < 10 required

The guard retains the previous plan

Repair 3 then passes with a 22-jump minimum in the proxy.

182 regressive proposals blocked

The controller preserves progress. Model proposals can still regress. No-ops are counted separately.

The evidence supports a narrow conclusion

Primary 4B comparison: inconclusive

Local feedback vs no diagnostic text
Brief-cluster p = 0.125

Secondary 9B comparison: exploratory

8 paired wins, 0 losses
Nominal brief-cluster p = 0.0156

Random repair remains competitive

14/24 vs 11/24 on 9B
Superiority not established, p = 0.375

Native-game evidence is informal

One participant reported completion after M01/M03 were supplied. Mode and attempts unknown.

Approximate physics and 6 known families. A held-out replication is needed.

What DashLab delivers

A working generation and repair pipeline

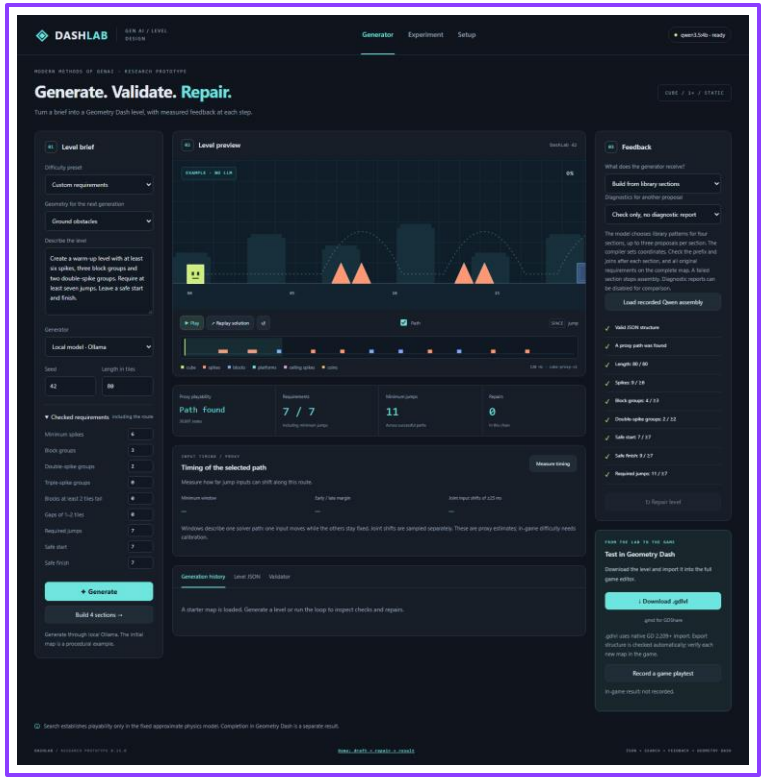
Local Qwen, independent checks, recorded replay and native Geometry Dash export.

A bounded research result

Diagnostic text helped 9B relative to the no-text repair control.

A reproducible next step

Retain all 288 chains, then preregister a held-out-family replication and systematic game playtests.



Questions

Results across six task families

Qwen 4B

Family	WN	WF	LN	LF	WR	LR
Intro	3/4	2/4	1/4	2/4	4/4	2/4
Blocks	1/4	0/4	0/4	3/4	1/4	3/4
Spikes	0/4	2/4	0/4	0/4	2/4	0/4
Stairs	0/4	0/4	0/4	1/4	0/4	2/4
Dense	0/4	0/4	0/4	0/4	0/4	1/4
Gauntlet	2/4	1/4	1/4	1/4	1/4	1/4

Qwen 9B

Family	WN	WF	LN	LF	WR	LR
Intro	1/4	3/4	1/4	3/4	3/4	3/4
Blocks	0/4	2/4	2/4	4/4	1/4	4/4
Spikes	1/4	3/4	2/4	3/4	3/4	2/4
Stairs	0/4	0/4	0/4	1/4	0/4	0/4
Dense	0/4	1/4	0/4	1/4	0/4	1/4
Gauntlet	1/4	1/4	1/4	2/4	1/4	1/4

W = whole plan, L = local section
N = no diagnostic text, F = feedback, R = random repair

Each cell contains 4 cases. These are new numeric briefs in existing families.

Paired comparisons and the primary endpoint

Comparison	Role	Wins / losses	Brief p	Family p
4B feedback vs no text	Primary	5 / 0	0.125	0.25
9B feedback vs no text	Secondary	8 / 0	0.015625	0.03125
9B feedback vs random	Secondary	4 / 1	0.375	0.25

24 matched cases per row. Seeds are grouped by brief.
9B p-values are nominal and exploratory after development history.

The primary endpoint stayed fixed. No general superiority over random repair is established.

Reproducibility, compute and diversity

Audit item	Recorded outcome
Offline replay	288 / 288 chains
Frozen source files	21
Actual model requests	527 completed, 0 API errors
9B local feedback diversity	13 geometries among 14 successes

Qwen3.5 4B and 9B, Q4_K_M, via Ollama
RTX 3060 Ti, 8 GB. Model profiles ran in serial batches.

Geometry Jaccard diversity ≈ 0.809. It measures object tokens, relative position and height, not fun.

Related work and our contribution

PCGML Summerville et al., 2017/2018	Representation and functional content
Self-Refine Madaan et al., 2023	Iterative feedback and revision
Self-Debugging Chen et al., 2023	Repair using program behavior
Limits of intrinsic self-correction Huang et al., 2023/2024	Motivation for external measurements

Our contribution: an exportable prototype and fully logged repair comparisons